Calculus 1 sec 2: Assignment 1, due Thursday Feb 5, 2026

Submit the completed assignment to https://www.gradescope.com. (which you access via Courseworks) Answer each question on a separate sheet, and please **match your answers with the correct questions on Gradescope**.

Only hand-written solutions are accepted.

- This assignment has four questions, worth a total of 15 points. It accounts for about 3% of your final grade. The late submission deadline (no penalty) is Friday, Friday Feb 6, 11:59 pm. Submissions after the late due date will not be accepted unless you write to Prof McDuff with a valid excuse.
- Collaboration is allowed, but you must write up your solutions independently and in your own words. Copying from others, publications, or websites (even with minor changes) will be treated as plagiarism.
- Neatness, clarity, and correct notation matter! Show all steps, use clear explanations, and write in complete sentences where needed. Graders will only evaluate what you write, so unclear or incomplete work will affect your grade, even with the correct answer.
- Attempt all questions (and parts), and clearly indicate which question you are answering.

1 (i) [1 pt] Write the function $f(x) = \left(\frac{x^2+1}{x^2-1}\right)^{1/3}$ as a composite of three functions.

(ii) [2 pt] Find the domain and range of f. Express your answer in interval notation.

2. (i) [2 pts] Find the range of $f(x) = 2 - 4|\cos(\pi x) - 1|$ algebraically.
Show all the steps. (Hint: start from the range of $\cos(x)$ *then* $\cos(\pi x)$ *then* $\cos(\pi x) - 1$ *and so on.*

(ii) [3pts] Graph the function $f(x) = 2 - 4|\cos(\pi x) + 1|$ by hand, not by plotting points, but by starting with the graph of $g(x) = \cos(x)$ and then applying the appropriate transformations as in part (i). So there should be 6 plots in all, including that of g. Label the graphs, and explain your work.

3. (i) [1 pt] A population $P(t)$ growing exponentially doubles after 5 years. Write down a formula for the size $P(t)$ of this population after t years. Your equation should have no unknown constants except for $P(0)$, its initial size.

(ii) [1 pt] If the population has size $10,000$ after 20 years, what is its initial size? Explain your reasoning.

4. [1 pt each] For (i)-(iii) use only the laws of exponents and logarithms and explain all steps in your calculation.

(i) Evaluate $\log_4(48) - \log_4(3)$.

(ii) Evaluate $10^{-\frac{1}{2}\log_{10} 49}$.

(iii) Explain why $\log_a(b) = 1/\log_b(a)$.

(iv) Calculate $\tan(\arcsin(-1/2))$ showing all steps.

(v) Show that the function $\frac{1-e^x}{1+e^x}$ is an odd function.